

Pandas Date Functions — Complete Guide

This document summarizes the most important date and time functions in Pandas, with explanations and examples.

1. Creating Date Ranges

- `pd.date_range(start=None, end=None, periods=None, freq='D')`
Example: `pd.date_range("2025-01-01", periods=5, freq="D")`

- `pd.period_range(start, end, freq='M')`
- `pd.timedelta_range(start='0s', periods=5, freq='1D')`

2. Converting to Datetime

- `pd.to_datetime(arg, format=None, errors='raise')`

3. Timestamps and Offsets

- `pd.Timestamp('2025-01-01')`
- `pd.Timedelta('7 days')`
- `pd.DateOffset(months=1, years=2)`

4. Extracting Date Parts with .dt

- `.dt.year`, `.dt.month`, `.dt.day`, `.dt.day_name()`, `.dt.quarter`, `.dt.is_leap_year`

5. Date Arithmetic

- Using `Timedelta`: `df["date"] + pd.Timedelta(days=7)`
- Using `DateOffset`: `df["date"] + pd.DateOffset(months=1)`

6. Resampling Time Series

- `ts.resample("W").sum()`
- `ts.resample("M").mean()`

7. Rolling & Expanding Windows

- `.rolling(window).mean()`
- `.expanding().mean()`
- `.ewm(span=3).mean()`

8. Shifting and Lagging

- `.shift()`
- `.pct_change()`

9. Timezones

- `.tz_localize("UTC")`
- `.tz_convert("Asia/Kolkata")`

10. Offsets for Year/Month/Quarter Boundaries

- `offsets.YearEnd`, `offsets.YearBegin`, `offsets.QuarterEnd`, `offsets.MonthBegin`

11. Special Constructors

- `pd.Period("2025-01", freq="M")`
- `pd.Interval(start, end)`

Practice Exercises

1. Generate all Fridays in 2025.
2. Renewal date = signup + 1 year.
3. Group sales by weekday.
4. Resample stock prices weekly.
5. Convert UTC → US/Eastern.
6. Fiscal year end in March.

Summary

- `pd.to_datetime` for conversion
- `Timedelta` for fixed intervals
- `DateOffset` for calendar shifts
- `.dt` accessor for components
- `resample`, `rolling`, `expanding`, `ewm` for time-series
- `tz_localize` and `tz_convert` for timezones